

# Instrumental Variables

PS200C, Spring 2026

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# Where we are

- For SOO, we leaned on **conditional ignorability**: find an  $X$  that closes all back-doors, then adjust for it.
- For DID, we traded ignorability in levels for ignorability in trends (**parallel-trends**)
- Today, a different bargain: Try to find an “encouragement” to treatment, which only matters through treatment, identifying an effect in a sub-group.

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This is the **intent-to-treat (ITT)** or **reduced form** effect of the card.
- **But it isn't the effect of pizza.** How do we get back to that?

# Backing it out: four kinds of people

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**Punchline:** the ITT over everyone will be an average over these and so will be a *watered-down* version of the pizza effect. Specifically,

# Backing it out: the algebra

Averaging the ITT over the four types:

$$\begin{aligned} \text{ITT} &= \underbrace{(\text{ITT} \mid \text{never-taker})}_{=0} \cdot \text{Pr}(\text{never-taker}) \\ &+ \underbrace{(\text{ITT} \mid \text{always-taker})}_{=0} \cdot \text{Pr}(\text{always-taker}) \\ &+ (\text{ITT} \mid \text{complier}) \cdot \text{Pr}(\text{complier}) \\ &+ \underbrace{(\text{ITT} \mid \text{defier})}_{=0, \text{ by assumption}} \cdot \text{Pr}(\text{defier}) \\ \Rightarrow \text{ITT} \mid \text{complier} &= \frac{\text{ITT}}{\text{Pr}(\text{complier})} \end{aligned}$$

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So we can recover an ITT-for-compliers.

And for compliers, the ITT is the ATE.



# From ITT-on-compliers to LATE

One more thing you need to know:  $\Pr(\text{complier})$  is the “first stage” effect:

$\Pr(\text{complier}) = \mathbb{E}[D_1 - D_0] = \mathbb{E}[D|Z = 1] - \mathbb{E}[D|Z = 0]$  (by ignorability of  $Z$ ),  
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Putting it together: We call the ATE among the compliers the “local average treatment effect” (LATE):

$$\text{LATE} = \frac{\text{ITT}}{\Pr(\text{complier})} = \frac{\mathbb{E}[Y | Z = 1] - \mathbb{E}[Y | Z = 0]}{\mathbb{E}[D | Z = 1] - \mathbb{E}[D | Z = 0]}$$

# Formalizing this potential-outcomes machinery

Let  $Z_i \in \{0, 1\}$  be the encouragement (*instrument*) and  $D_i \in \{0, 1\}$  the treatment.

**Potential treatments.**  $D_{1i}, D_{0i}$ : would you take the treatment if encouraged ( $Z = 1$ ) or not ( $Z = 0$ ). We see  $D_i = Z_i D_{1i} + (1 - Z_i) D_{0i}$ .

**Four principal strata**, defined by  $(D_{0i}, D_{1i})$ :

|              | $D_{0i} = 0$    | $D_{0i} = 1$  |
|--------------|-----------------|---------------|
| $D_{1i} = 0$ | never-taker     | <b>defier</b> |
| $D_{1i} = 1$ | <b>complier</b> | always-taker  |

**Monotonicity**  $D_{1i} \geq D_{0i}$  for all  $i$  rules out the red cell.

# The assumptions

**1. Exclusion:**  $Y(d, z) = Y_d$ :  $Z$  affects  $Y$  only through  $D$ .

- **Not implied** by  $Z$  being random:  $Z$  can be random but *still* affect  $Y$  through some channel other than  $D$ .
- Not testable though sometimes falsifiable.

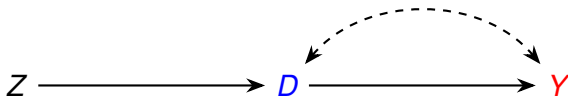
**2. Ignorability** of  $Z$ :  $\{Y(d), D(z)\} \perp\!\!\!\perp Z$  for all  $d, z$ .

- Ideally  $Z$  is literally randomized, or pseudo-randomized (some “shock”)

**3. First stage / relevance:**  $\Pr(D_1 = 1) \neq \Pr(D_0 = 1)$  — i.e. there are some compliers. Testable: regress  $D$  on  $Z$ .

**4. Monotonicity:**  $D_{1i} \geq D_{0i}$  — no defiers. Not testable, but often qualitatively defensible (e.g. a lottery winner is no *less* likely to enroll).

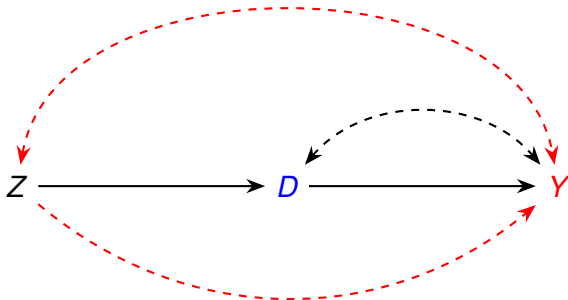
# The IV DAG



This would be good.

- Unobserved confounders of  $D$  and  $Y$  – no problem!
- But what's important is what's not on here...

# The IV DAG: edges that should *not* be there



Red edges are the ones an IV story *must* rule out:

- Top:  $Z \leftrightarrow Y$ : any unobserved factor associated with both  $Z$  and  $Y$ . **Violates ignorability of  $Z$ .**
- Bottom:  $Z \rightarrow Y$ : a direct path from  $Z$  to  $Y$  not via  $D$ . **Violates exclusion.**



# Estimation: Wald and 2SLS

**Wald estimator** (sample analog of the ratio):

$$\hat{\beta}_{\text{Wald}} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}$$

**Two-stage least squares (2SLS):**

- 1 First stage:  $D_i = \pi_0 + \pi Z_i + \eta_i$ . Compute  $\hat{D}_i$ .
- 2 Second stage:  $Y_i = \beta_0 + \beta \hat{D}_i + \varepsilon_i$ .

With one binary  $Z$  and one binary  $D$ ,  $\hat{\beta}_{2\text{SLS}} = \hat{\beta}_{\text{Wald}}$ .

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In practice: for  $\mathbf{X} = [1 \ D]$ ,  $\hat{\beta}_{IV} = (Z^\top \mathbf{X})^{-1} Z^\top \mathbf{Y}$ .

- In R: `ivreg(Y ~ D | Z)` from the AER package.

# Estimation: covariates and weak instruments

**Adding covariates** when ignorability/exclusion only hold conditional on  $X$ :

- Add  $X$  to *both* stages.
- With heterogeneous effects, this no longer cleanly identifies a LATE; the complier set varies with  $X$ .

**Weak instrument diagnostics:**

- First-stage  $F$ -statistic; rule of thumb  $F > 10$ .
- Better: [Anderson–Rubin](#) confidence intervals — valid even when the instrument is weak.

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- Easy, unbiased, but **understates the effect of the treatment itself**.

**The IV “correction”:** divide that ITT by compliance rate to get back the “undiluted” effect of actually *taking* the treatment, but only for *compliers*.

- **Ignorability** of  $Z$ : holds *by design* (the experimenter randomized).
- **Exclusion**: could still be violated but could be defensible – some  $Y$ s are hard to move without special treatment.
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This is standard fare in many RCTs and field experiments.



# Three famous IV Examples

| Setup                                                                          | Instrument $Z$                              | Treatment $D$                 | Outcome $Y$          |
|--------------------------------------------------------------------------------|---------------------------------------------|-------------------------------|----------------------|
| <b>Acemoglu, Johnson &amp; Robinson (2001)</b><br><i>settler mortality</i>     | European settler mortality,<br>c. 1500–1900 | quality of institutions today | log GDP per capita   |
| <b>Angrist &amp; Krueger (1991)</b><br><i>quarter of birth</i>                 | quarter of birth (Q1 vs. Q3/Q4)             | years of schooling            | log earnings         |
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Let's worry about exclusion in each.

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**Angrist's idea:** use the [Vietnam-era draft lottery](#) (1969–72). Each birthday was randomly assigned a number 1–365; the Selective Service then announced a yearly cutoff, and men with numbers *below* the cutoff were called for induction.

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**Critical:**  $Z \neq D$ . Among the called ( $Z = 1$ ), only  $\sim 35\%$  served (deferments, failed physicals). Among the not-called ( $Z = 0$ ),  $\sim 19\%$  still served (volunteers). First stage  $\approx 0.16$ .



# Wald estimates from Angrist (1990): white men, 1950 cohort

|             |                                                       |                                         |                           |
|-------------|-------------------------------------------------------|-----------------------------------------|---------------------------|
| ITT         | $(\bar{Y}^e - \bar{Y}^n, 1981 \text{ FICA earnings})$ | -\$435.8                                | (s.e. 210.5)              |
| First stage | $(\hat{p}^e - \hat{p}^n, \text{veteran rate})$        | 0.159                                   | (s.e. 0.040)              |
| Wald        | = ITT / first stage                                   | $\frac{-435.8}{0.159} \approx -\$2,741$ | (s.e. $\approx \$1,335$ ) |

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What's the interpretation of this result?

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**LATE for whom?** Lottery-induced compliers: those who served *because* of their number. Not always-takers (volunteers), not never-takers (medically deferred).



# Rules of the game

- 1 Show **Relevance**:  $Z$  moves  $D$ . *Look at the first stage.*
- 2 **Ignorability + exclusion**: ignorability is given by random  $Z$ ; still worry about exclusion. *Not verifiable but explain why you believe it. You can show certain (observed) channels can be ruled out.*
- 3 **Monotonicity**: no defiers. *Often you can argue this on qualitative grounds.*
- 4 **Interpretation**: LATE for compliers. *Who are the compliers? Are they policy-relevant?*